BEE 4750 Homework 4: Linear Programming and Capacity Expansion

Name: Parker Catena, Regina Velasco

ID: pdc76, rv297

Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750 /HW 4/hw4-parker-regina`
   Installed ArgCheck      v2.5.0
   Installed MarkdownTables  v1.1.0
   Installed DisplayAs     v0.1.6
   Installed StatsBase     v0.34.7
   Installed BenchmarkTools  v1.6.3
   Installed DataFrames     v1.8.1
   Installed PrettyTables    v3.1.0
   Installed JuMP        v1.29.2
Precompiling project...
    520.5 ms   StaticArraysCore
    683.0 ms   ArgCheck
    546.0 ms   CommonSubexpressions
    691.1 ms   DisplayAs
    714.1 ms   StructUtils
    799.4 ms   StructTypes
   1027.2 ms   IrrationalConstants
    514.0 ms   LoggingExtras
    533.7 ms   Compat
    681.6 ms   libaom_jll
    598.8 ms   Expat_jll
    727.5 ms   OpenSSL_jll
    657.6 ms   CodecBzip2
    889.7 ms   HiGHS_jll
    985.1 ms   DiffResults
   1190.4 ms   StructUtils → StructUtilsTablesExt
   1743.3 ms   MarkdownTables
   1357.2 ms   LogExpFunctions
   1204.6 ms   Compat → CompatLinearAlgebraExt
   1149.5 ms   Fontconfig_jll
   3392.4 ms   JSON
   5261.8 ms   MutableArithmetics
   2139.2 ms   OpenSSL
   2764.4 ms   StatsBase
   2449.9 ms   SpecialFunctions
   1169.2 ms   Cairo_jll
   2276.3 ms   Qt6Base_jll
   1054.9 ms   DiffRules
   2175.8 ms   BenchmarkTools
   1431.1 ms   ColorVectorSpace → SpecialFunctionsExt
   1209.4 ms   HarfBuzz_jll
   7205.1 ms   JSON3
   1361.3 ms   Qt6ShaderTools_jll
```

```
 1272.1 ms    libass_jll
 1276.2 ms    Pango_jll
 1665.6 ms    Qt6Declarative_jll
  393.9 ms    Qt6Wayland_jll
 1614.9 ms    FFMPEG_jll
 3972.9 ms    ForwardDiff
  960.1 ms    FFMPEG
 1387.7 ms    GR_jll
13771.9 ms    HTTP
 2843.4 ms    GR
24502.1 ms    PrettyTables
31447.0 ms    MathOptInterface
 2015.6 ms    MathOptIIS
 8517.1 ms    HiGHS
31780.1 ms    DataFrames
14475.4 ms    JuMP
 2228.3 ms    Latexify → DataFramesExt
44736.9 ms    Plots
51 dependencies successfully precompiled in 69 seconds. 171 already precompiled.
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

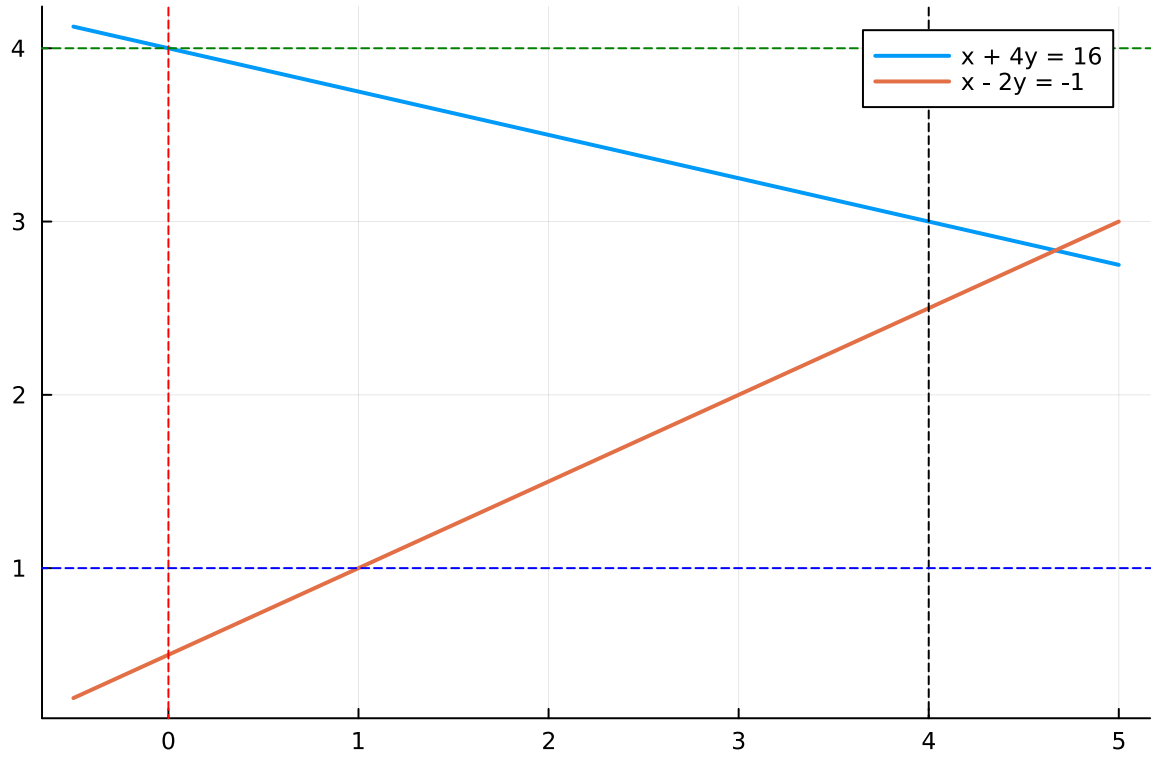
Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

```
#Set x range for plotting
x = -0.5:0.01:5.0

#Constraint functions
y1(x) = (16 - x) / 4
y2(x) = (x + 1) / 2

#Plot constraint functions across all x values
plt = plot(x, y1.(x), lw=2, label="x + 4y = 16")
plot!(plt, x, y2.(x), lw=2, label="x - 2y = -1")

#Draw the vertical and horizontal lines representing
#the bounds for constraints
vline!([0], linestyle=:dash, color=:red, label="")
vline!([4], linestyle=:dash, color=:black, label="")
hline!([1], linestyle=:dash, color=:blue, label="")
hline!([4], linestyle=:dash, color=:green, label="")
```



Using the above graph of constraints, we know that the maximum value must occur at an intersection of the boundaries. We can use these intersections to check the values for the

maximum of the function $3x+2y$. The feasible intersections, as determined by are as follows:

$$(0,1),(0,4),(1,1),(4,\frac{5}{2}),(4,3)$$

We can then check these pairs with the function $3x+2y$:

$$3(0)+2(1)=2, \quad (0,1)$$

$$3(0)+2(4)=8, \quad (0,4)$$

$$3(1)+2(1)=5, \quad (1,1)$$

$$3(4)+2(2.5)=17, \quad (4,2.5)$$

$$3(4)+2(3)=18, \quad (4,3)$$

Therefore, we can see that at $(4,3)$ we achieve the maximum value of $3x+2y=18$.

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36

Crop	Production Cost ($/ha-yr)	Selling Price ($/kg)
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
#Problem 2.1
using JuMP # optimization modeling syntax
using HiGHS # optimization solver

crop_model = Model(HiGHS.Optimizer) # initialize model object
```

```
@variable(crop_model, s[0:2] >= 0) # non-neg constraints for soy
@variable(crop_model, w[0:2] >= 0) # non-neg constraitns for wheat
@variable(crop_model, c[0:2] >= 0) # non-neg constraitns for corn


# index: 1 = soy, 2 = wheat, 3 = corn

# set up variables:
# selling prices ($/kg)
ps, pw, pc = 0.36, 0.27, 0.22
# production cost ($/ha)
cs, cw, cc = 350, 280, 390
# pesticide cost ($ per kg/ha applied)
pest = 70
# yields (kg/ha)
ys0, ys1, ys2 = 2900, 3800, 4400
yw0, yw1, yw2 = 3500, 4100, 4200
yc0, yc1, yc2 = 5900, 6700, 7900
# average rate caps (kg/ha)
Ls, Lw, Lc = 0.8, 0.7, 0.6
# land & demand
land_cap = 130
demand  = 250000 #for each crop

# OBJ: maximize profits over 130 total ha while remaining in regulatory
# compliance if demand for each crop (which is the maximum the market
# would buy) this year is 250,000 kg
# Objective = revenue - production costs - pesticide costs
@objective(crop_model, Max,
    # revenue
    + ps*(ys0*s[0] + ys1*s[1] + ys2*s[2])
    + pw*(yw0*w[0] + yw1*w[1] + yw2*w[2])
    + pc*(yc0*c[0] + yc1*c[1] + yc2*c[2])
    # production costs
    - cs*sum(s[i] for i in 0:2)
    - cw*sum(w[i] for i in 0:2)
    - cc*sum(c[i] for i in 0:2)
    # pesticide costs (rate * pest * hectares)
    - pest*( 0*s[0] + 1*s[1] + 2*s[2]
           + 0*w[0] + 1*w[1] + 2*w[2]
           + 0*c[0] + 1*c[1] + 2*c[2])
)
```

```
# Constraints

# Land
@constraint(crop_model, land, sum(s[i] for i in 0:2) + sum(w[i] for i in 0:2) + sum(c[i] for

# Demand caps (kg)
@constraint(crop_model, ds, ys0*s[0] + ys1*s[1] + ys2*s[2] <= demand)   # soy
@constraint(crop_model, dw, yw0*w[0] + yw1*w[1] + yw2*w[2] <= demand)   # wheat
@constraint(crop_model, dc, yc0*c[0] + yc1*c[1] + yc2*c[2] <= demand)   # corn

# Average pesticide caps: sum(rate*x)  L * sum(x)
@constraint(crop_model, pest_s, (0*s[0] + 1*s[1] + 2*s[2]) <= Ls*(sum(s[i] for i in 0:2)))
@constraint(crop_model, pest_w, (0*w[0] + 1*w[1] + 2*w[2]) <= Lw*(sum(w[i] for i in 0:2)))
@constraint(crop_model, pest_c, (0*c[0] + 1*c[1] + 2*c[2]) <= Lc*(sum(c[i] for i in 0:2)))  #


print(crop_model)
```

```
Max 694 s[0] + 948 s[1] + 1094 s[2] + 665.0000000000001 w[0] + 757 w[1] + 714 w[2] + 908 c[0]
Subject to
 land : s[0] + s[1] + s[2] + w[0] + w[1] + w[2] + c[0] + c[1] + c[2]  130
 ds : 2900 s[0] + 3800 s[1] + 4400 s[2]  250000
 dw : 3500 w[0] + 4100 w[1] + 4200 w[2]  250000
 dc : 5900 c[0] + 6700 c[1] + 7900 c[2]  250000
 pest_s : -0.8 s[0] + 0.19999999999999996 s[1] + 1.2 s[2]  0
 pest_w : -0.7 w[0] + 0.30000000000000004 w[1] + 1.3 w[2]  0
 pest_c : -0.6 c[0] + 0.4 c[1] + 1.4 c[2]  0
 s[0]  0
 s[1]  0
 s[2]  0
 w[0]  0
 w[1]  0
 w[2]  0
 c[0]  0
 c[1]  0
 c[2]  0
```

```
optimize!(crop_model)

println("Optimized soybean values:")
@show value.(s[i] for i in 0:2);
```

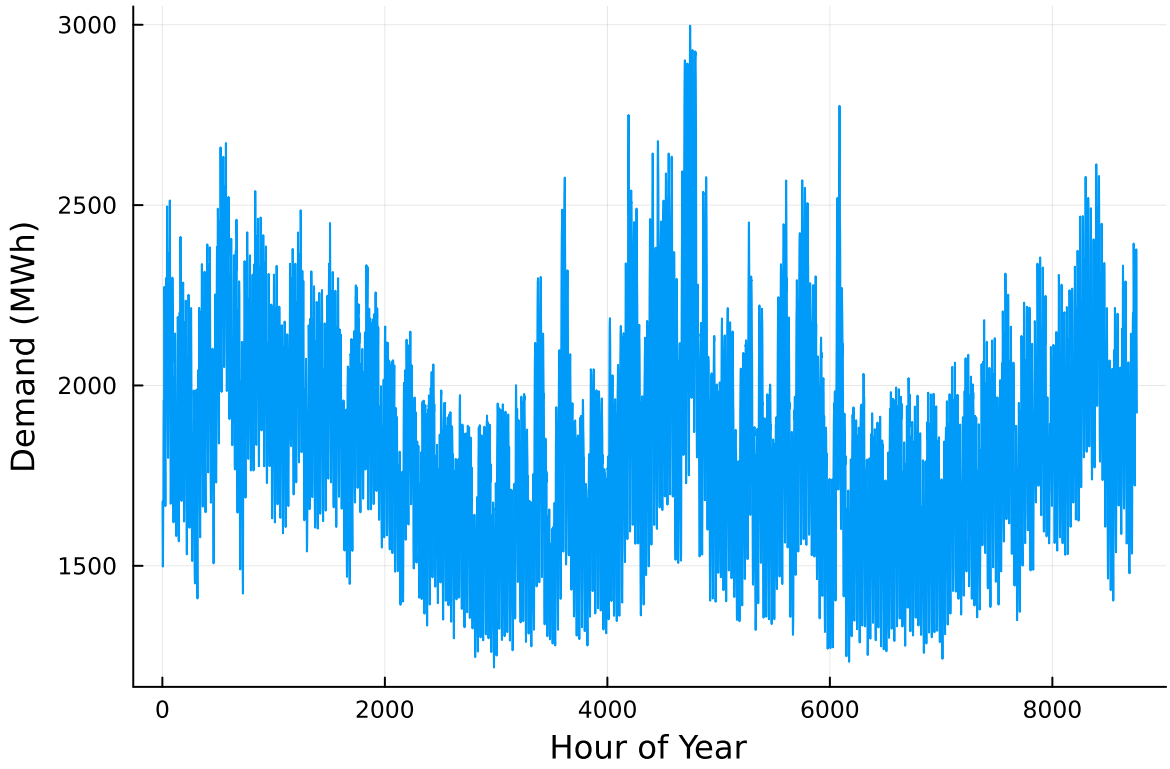
```julia
println("Optimized wheat values:")
@show value.(w[i] for i in 0:2);

println("Optimized corn values:")
@show value.(c[i] for i in 0:2);

println("Optimized objective value:")
@show objective_value(crop_model);
```

```
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP   has 7 rows; 9 cols; 27 nonzeros
Coefficient ranges:
  Matrix [2e-01, 8e+03]
  Cost   [7e+02, 1e+03]
  Bound  [0e+00, 0e+00]
  RHS    [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros  0s
7 rows, 9 cols, 27 nonzeros  0s
Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using EKK dual simplex solver - serial
  Iteration        Objective     Infeasibilities num(sum)
          0    -1.8170297740e+02 Ph1: 7(28.3601); Du: 9(181.703) 0s
          7     1.1674116702e+05 Pr: 0(0) 0s
Model status        : Optimal
Simplex   iterations: 7
Objective value     :  1.1674116702e+05
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
Optimized soybean values:
value.((s[i] for i = 0:2)) = [13.812154696132604, 55.24861878453038, 0.0]
Optimized wheat values:
value.((w[i] for i = 0:2)) = [6.743306417339563, 15.734381640458977, 0.0]
Optimized corn values:
value.((c[i] for i = 0:2)) = [26.92307692307692, 0.0, 11.538461538461547]
Optimized objective value:
objective_value(crop_model) = 116741.16702082448
```

```
println("Shadow values:")
@show shadow_price(land);

@show shadow_price(ds);
@show shadow_price(dw);
@show shadow_price(dc);

@show shadow_price(pest_w);
@show shadow_price(pest_w);
@show shadow_price(pest_c);
```

```
Shadow values:
shadow_price(land) = 729.4
shadow_price(ds) = 0.046353591160220996
shadow_price(dw) = -0.0
shadow_price(dc) = 0.04132307692307693
shadow_price(pest_w) = 91.99999999999993
shadow_price(pest_w) = 91.99999999999993
shadow_price(pest_c) = 108.67692307692309
```

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs ($/MW installed), variable costs ($/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

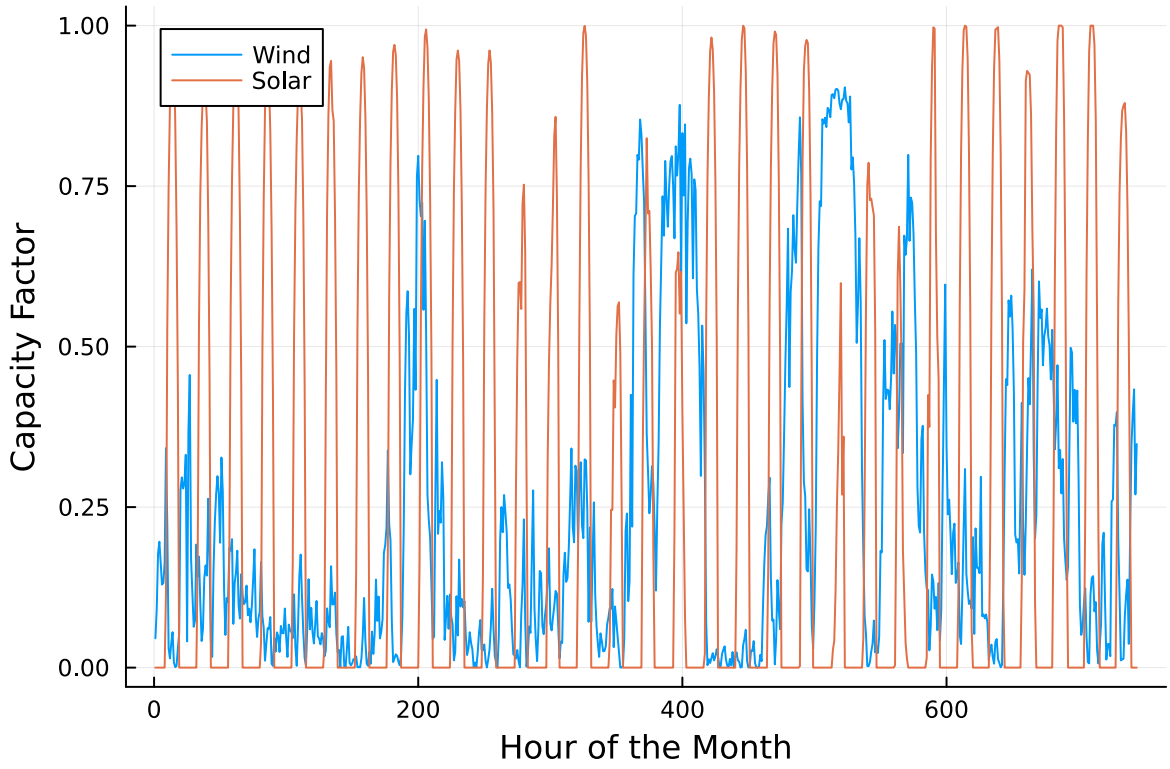
While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

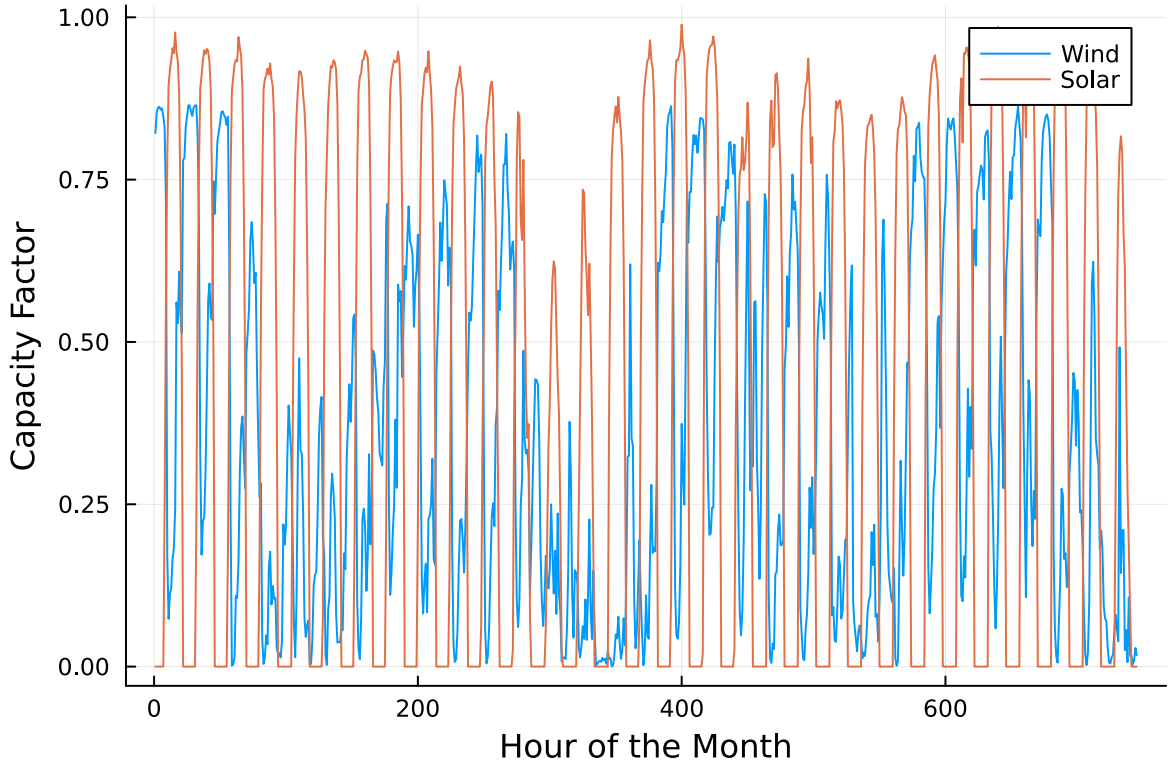
Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?


Wind
Solar
1.00
0.75
0.50
0.25
0.00
Capacity Factor
0
200
400
600
Hour of the Month


Wind
Solar
1.00
0.75
0.50
0.25
0.00
Capacity Factor
0
200
400
600
Hour of the Month

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

References

List any external references consulted, including classmates.